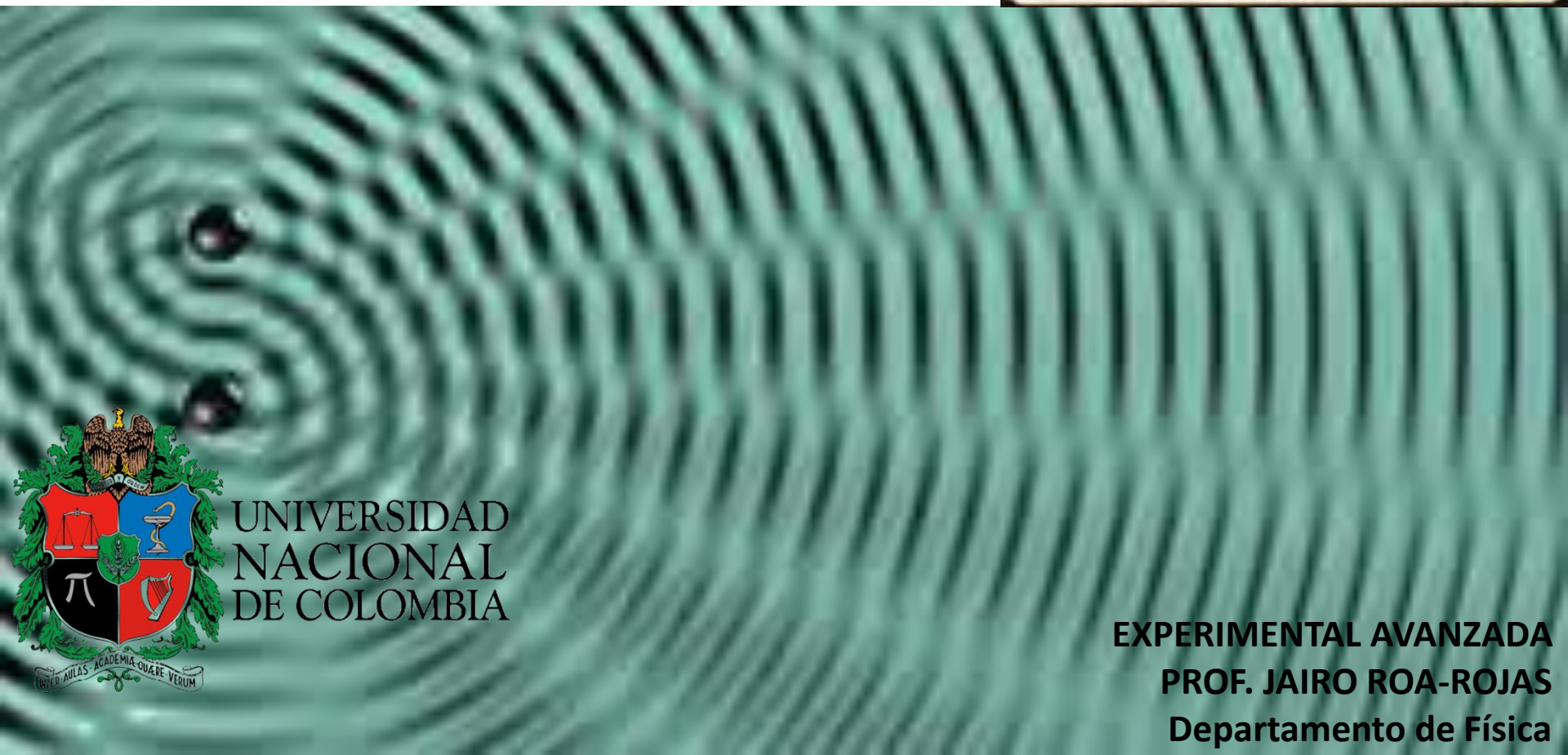
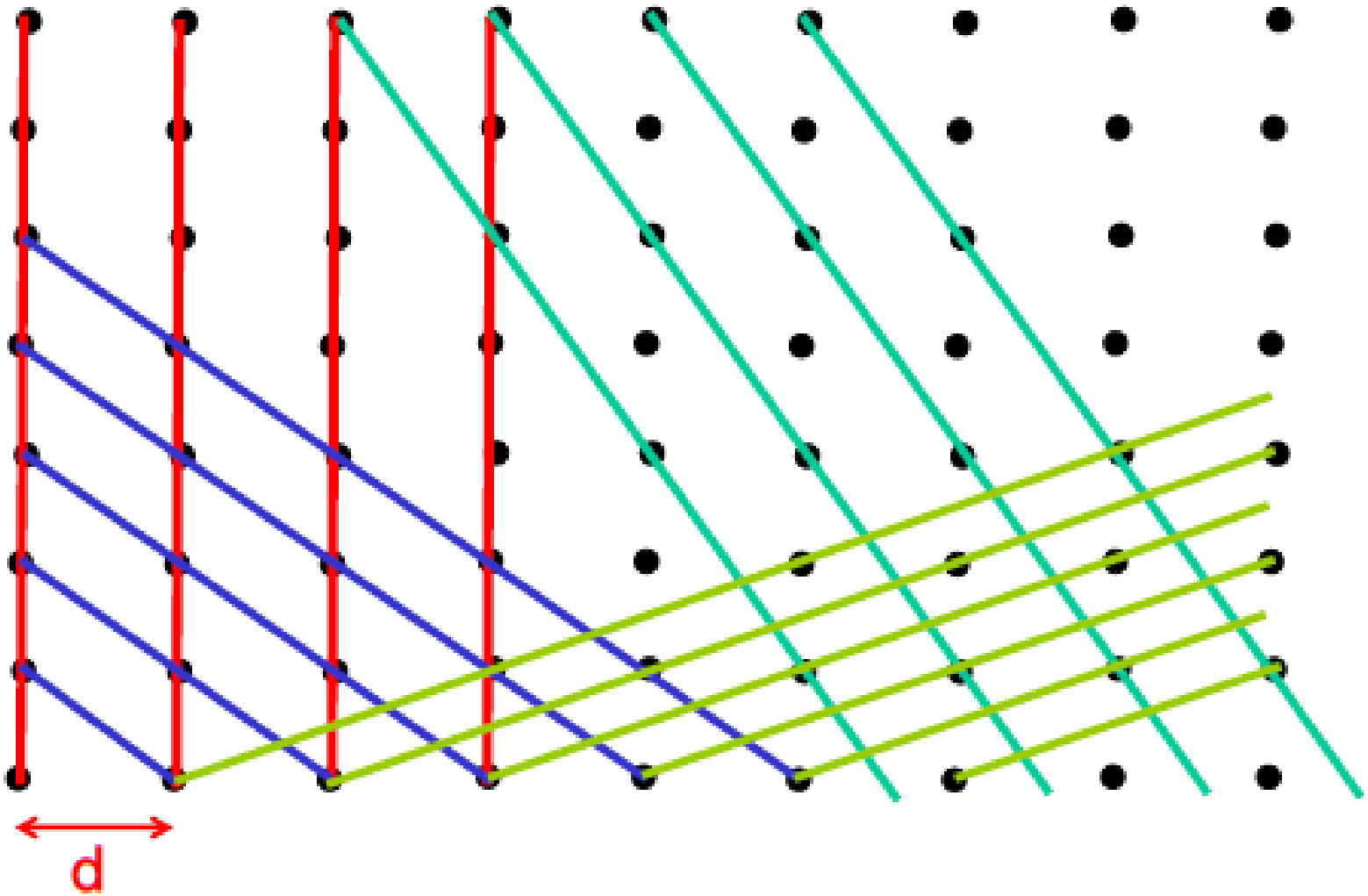
DIFRACCIÓN DE RAYOS-X Y RED RECÍPROCA



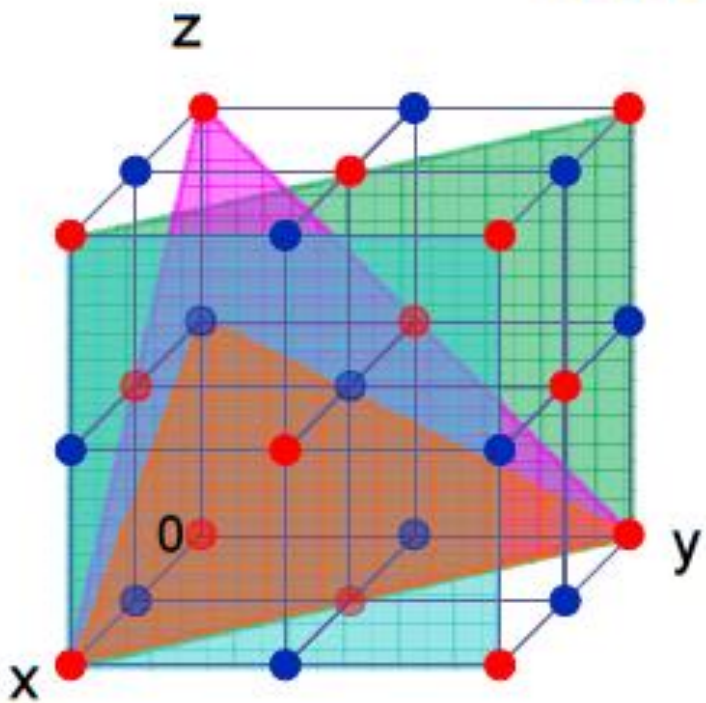
UNIVERSIDAD NACIONAL DE COLOMBIA

EXPERIMENTAL AVANZADA
PROF. JAIRO ROA-ROJAS
Departamento de Física

Planos y direcciones



Indices de Miller



Intersecciones a $(1,1,\infty)$
Índices de Miller (110)

Intersections at $(1,\infty,\infty)$
Miller indices (100)

Intersections at $(1,1,1)$
Miller indices (111)

Intersecciones a $(1,1,1/2)$
Índices de Miller (112)

Indices de Miller:

1. Encontrar intersecciones con ejes de celda unidad.
2. Tomar recíprocos
3. Encontrar enteros (más pequeños) con la misma razón

() Planos
{} Planos Equivalentes
Cubico
{100}: (100),(200),(001)
[] direcciones:
dirección x [100]

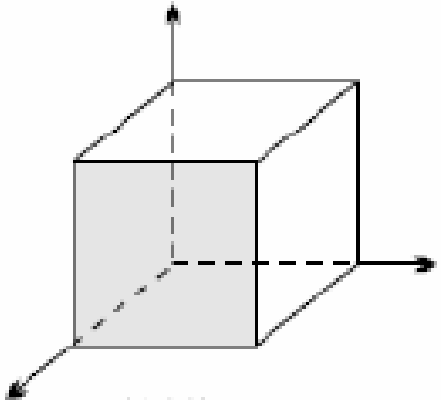
Distancia entre Planos

- Para una celda unidad con ejes ortogonales (cúbica, tetragonal, ortorrómbica) :

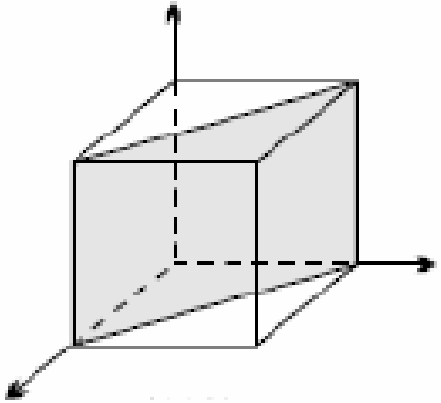
$$\frac{1}{d_{hkl}^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

- Para una celda unidad hexagonal:

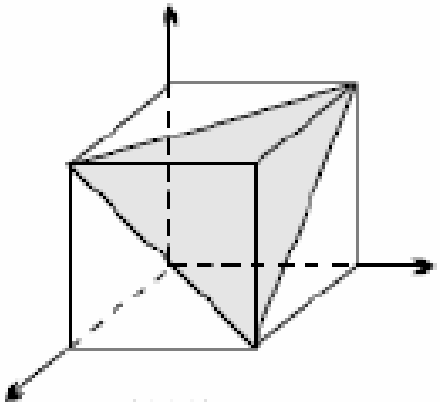
$$\frac{1}{d_{hkl}^2} = \frac{4\left(h^2 + k^2 + hk\right)}{3a^2} + \frac{l^2}{c^2}$$



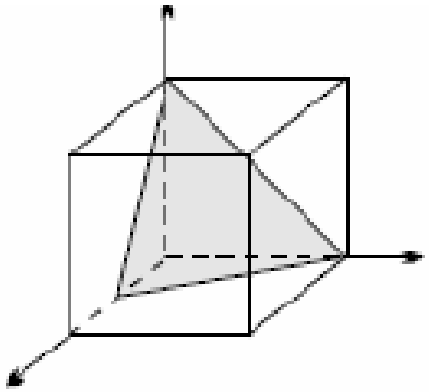
(100)



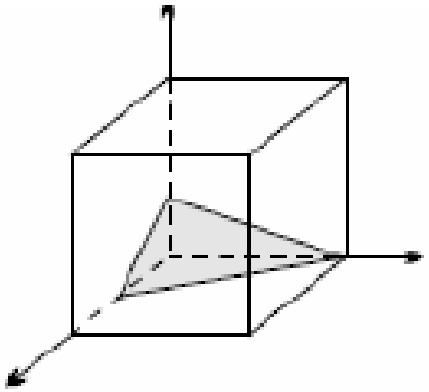
(110)



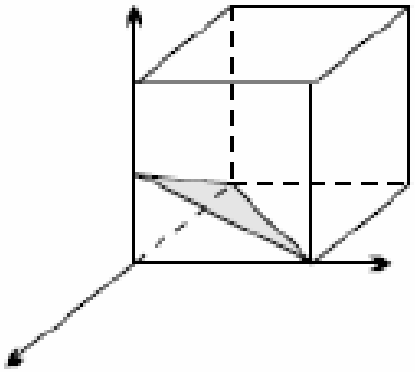
(111)



(211)

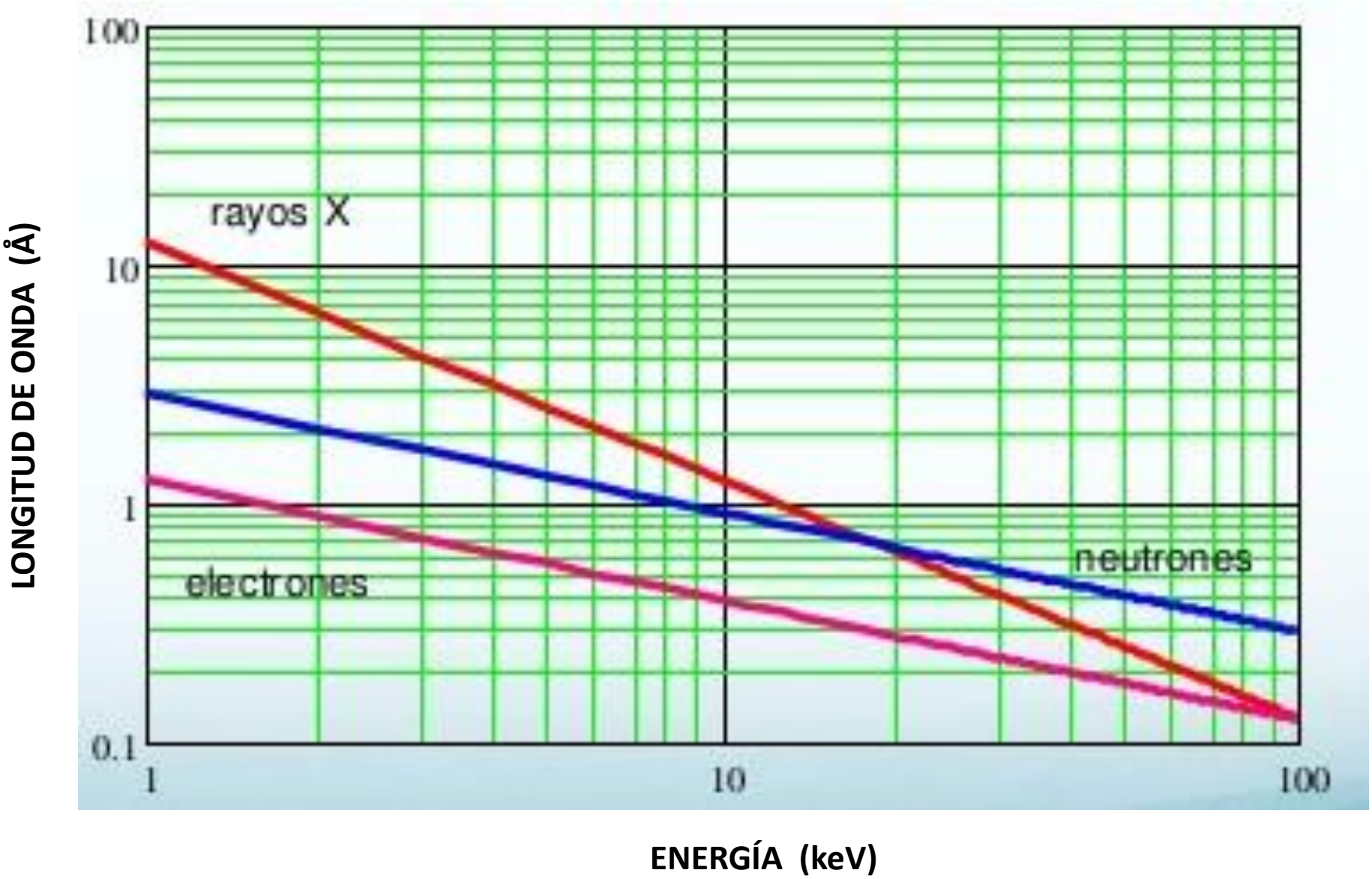


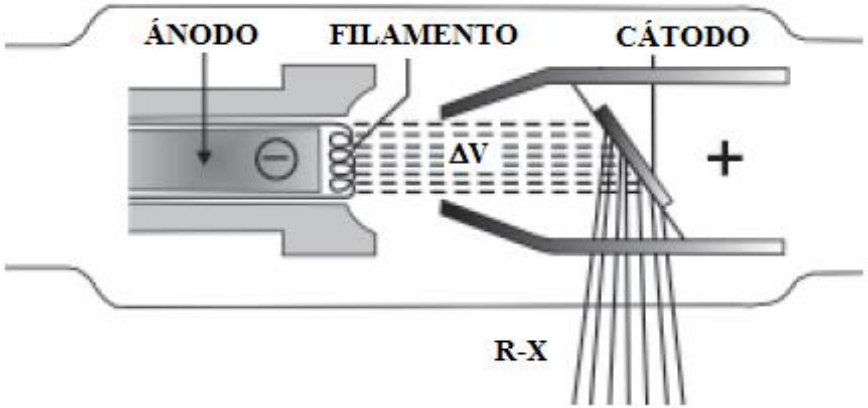
(213)

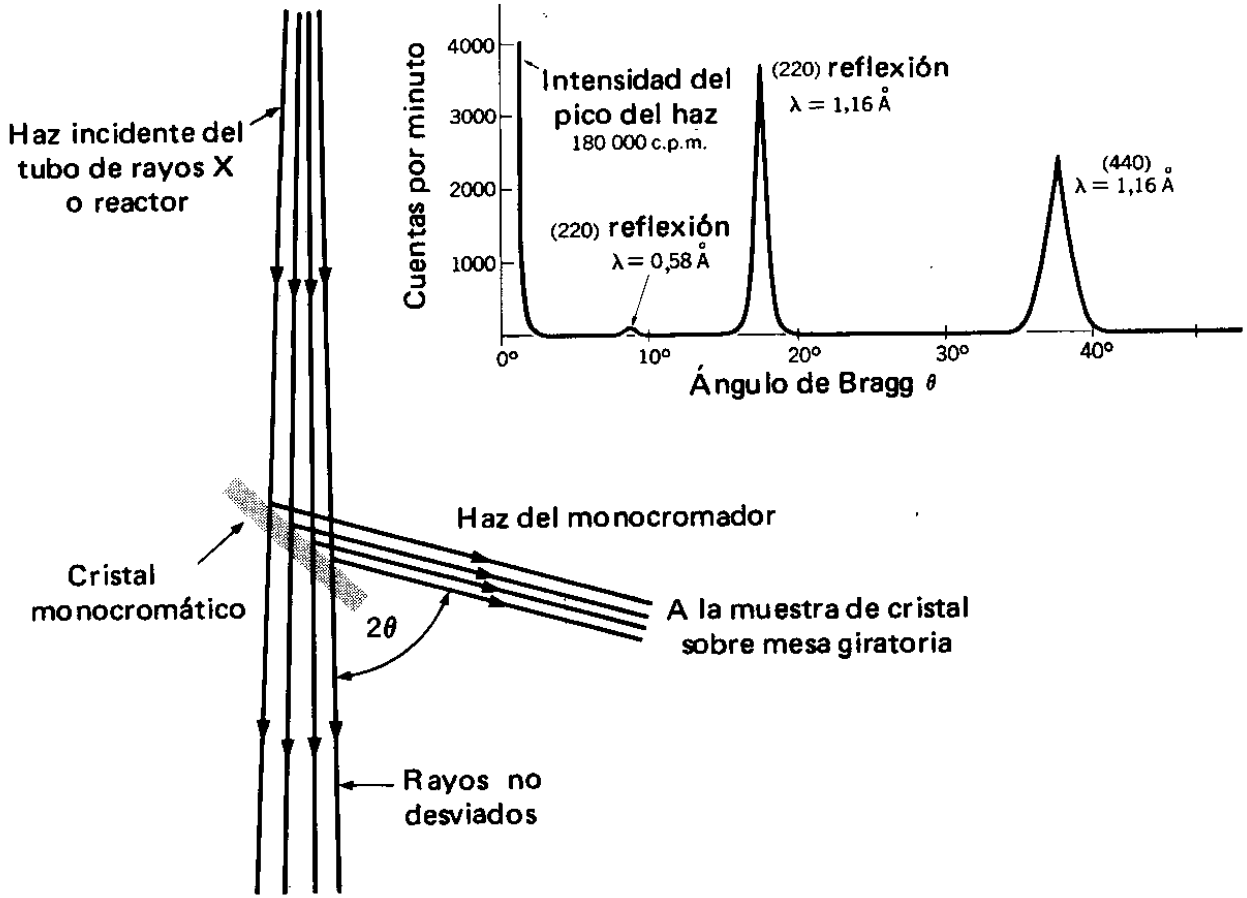


($\bar{1}12$)

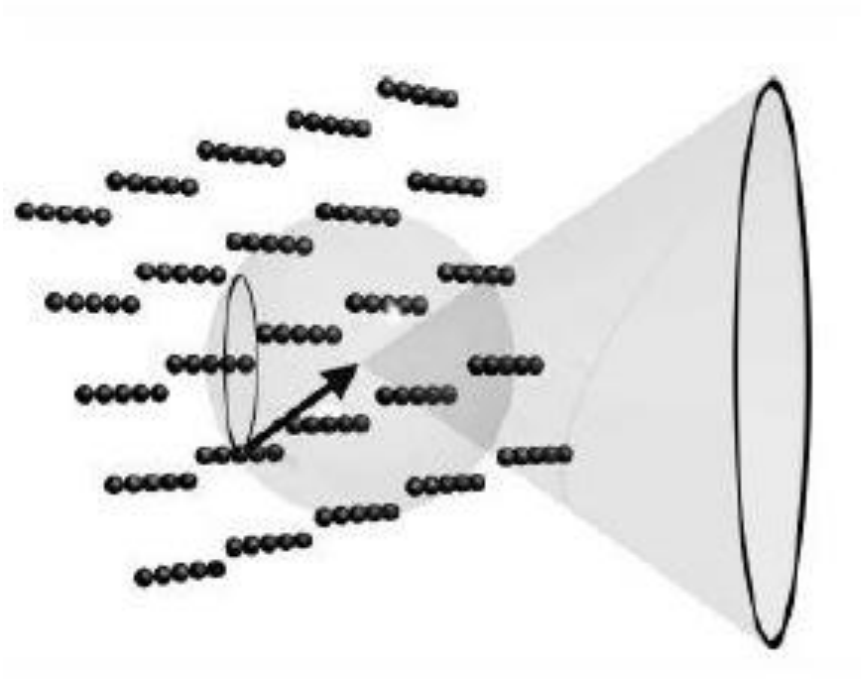
DIFRACCIÓN DE RAYOS X

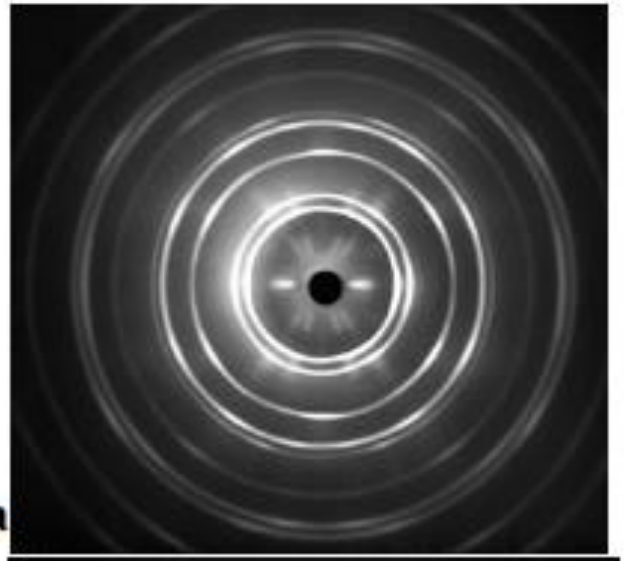



ÁNODO
FILAMENTO
CÁTODO
ΔV
+
R-X

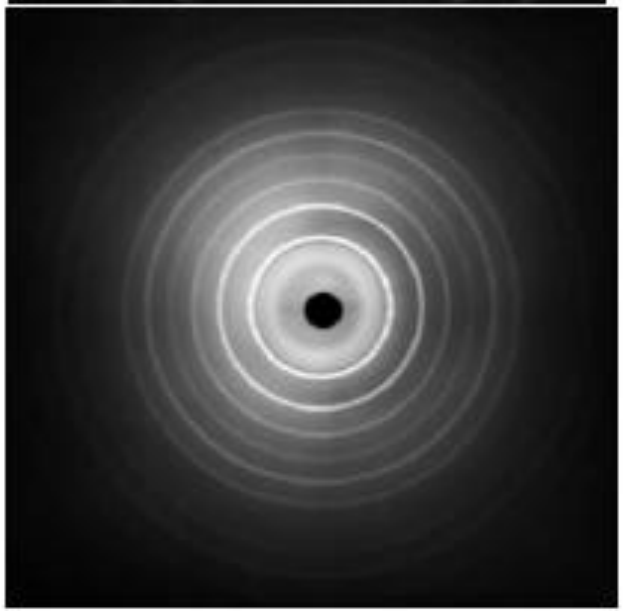

Haz incidente del tubo de rayos X o reactor
Cuentas por minuto
4000
3000
2000
1000
Intensidad del pico del haz
180 000 c.p.m.
(220) reflexión
λ = 0,58 Å
(220) reflexión
λ = 1,16 Å
(440)
λ = 1,16 Å
0°
10°
20°
30°
40°
Ángulo de Bragg θ
Haz del monocromador
Cristal monocromático
2θ
A la muestra de cristal sobre mesa giratoria
Rayos no desviados

Método de polvo (debye-Scherrer)





alumina



NaCl

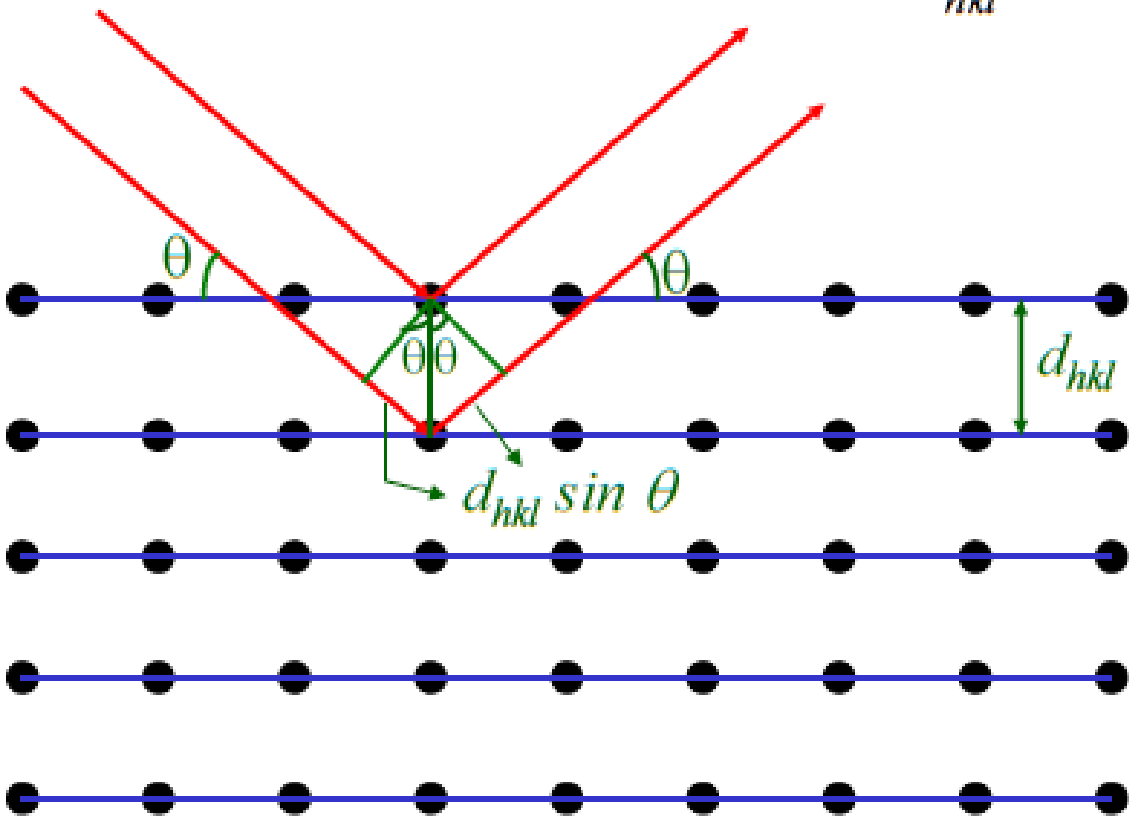
Taller:

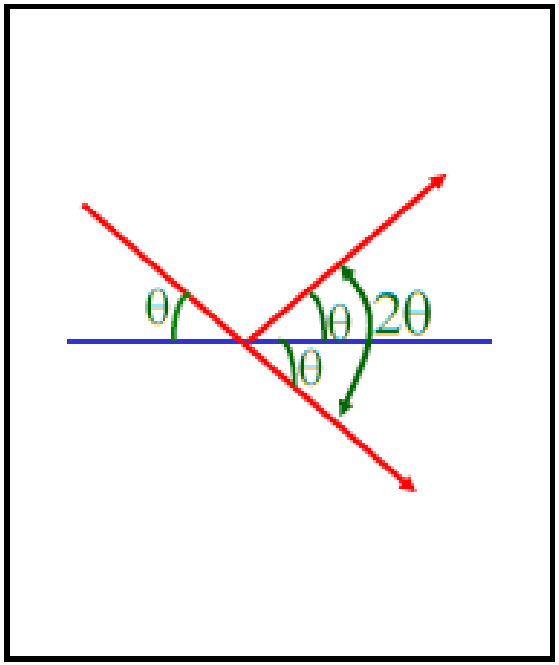
Investigue los métodos:

- De Laue
- Del cristal rotativo

Formulación de Bragg

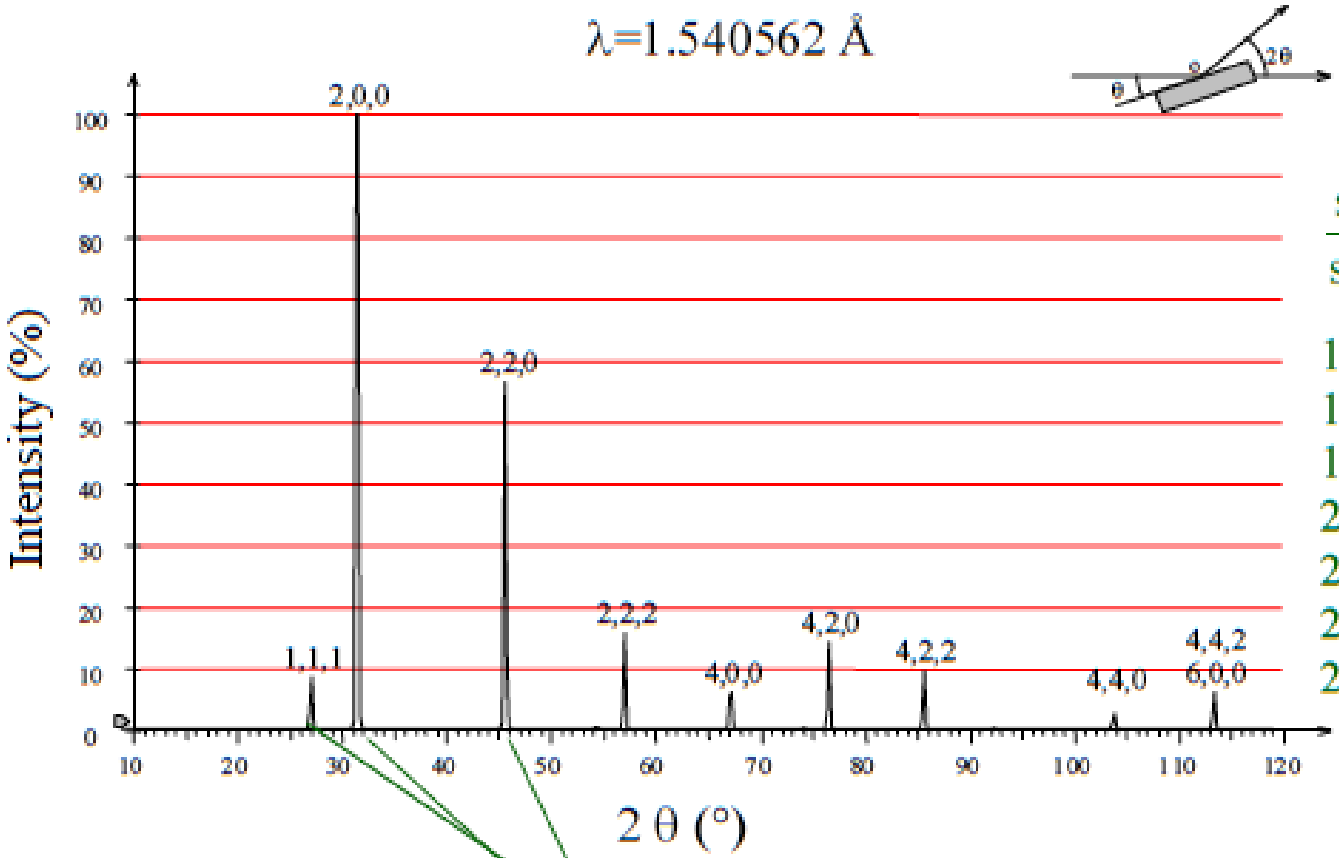
$$2d_{hkl}\sin\theta = n\lambda, \quad n = 1,2,3,\ldots$$





Existen reflexiones de Bragg solo para $\lambda \leq 2\ d_{hkl}$

Difracción de Bragg del NaCl



$$\frac{\sin^2\theta}{\sin^2\theta_i} = \frac{h^2+k^2+l^2}{h_i^2+k_i^2+l_i^2}$$

$100 \rightarrow 1$
$110 \rightarrow 2$
$111 \rightarrow 3$
$200 \rightarrow 4$
$210 \rightarrow 5$
$211 \rightarrow 6$
$220 \rightarrow 8$

$1.33\hat{3}$ (1.312)

$2.66\hat{6}$ (2.666)

$$2\sin\theta = \frac{\lambda\sqrt{h^2+k^2+l^2}}{a}$$

$$2\theta = 27.366^\circ$$
$$2\theta = 31.444^\circ$$
$$2\theta = 45.444^\circ$$

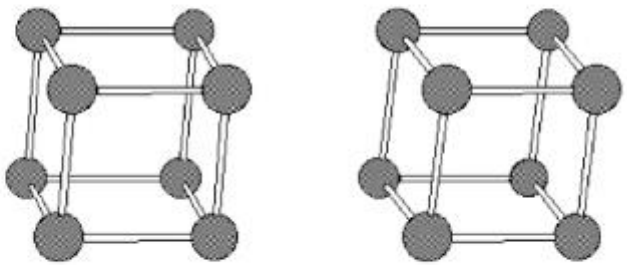
$$\Rightarrow \quad a_{NaCl} = 5.64\ \text{Å}$$

$$\vec{b}_1 = 2\pi \frac{\vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

$$\vec{b}_2 = 2\pi \frac{\vec{a}_3 \times \vec{a}_1}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

$$\vec{b}_3 = 2\pi \frac{\vec{a}_1 \times \vec{a}_2}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

Red recíproca de la red sc



$$\mathbf{a}_1 = a\mathbf{x} \qquad \mathbf{a}_2 = a\mathbf{x} \qquad \mathbf{a}_3 = a\mathbf{x}$$

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3}$$

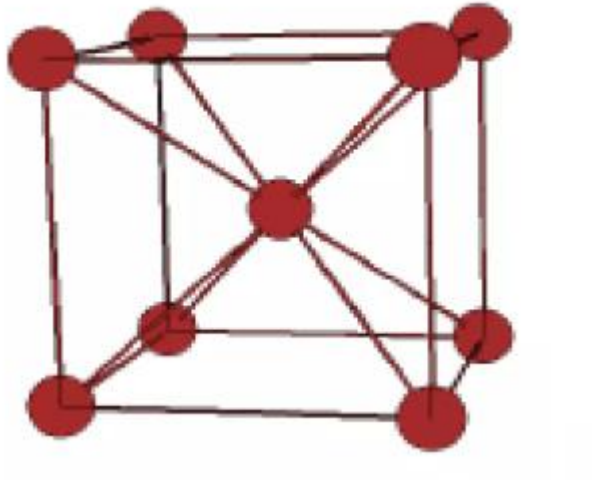
$$\mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3}$$

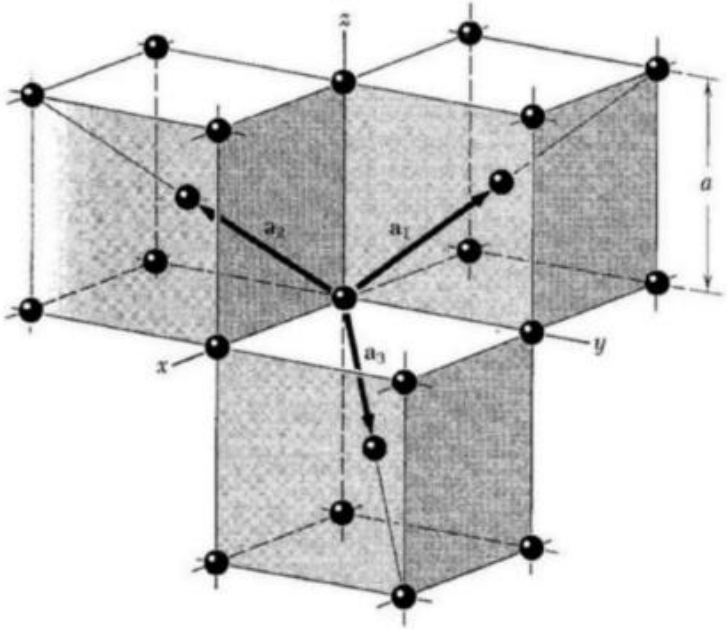
$$\mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3}$$

Los vectores de traslación de la red recíproca:

$$\mathbf{b}_1 = (2\pi / a)\mathbf{x} \qquad \mathbf{b}_2 = (2\pi / a)\mathbf{y} \qquad \mathbf{b}_3 = (2\pi / a)\mathbf{z}$$

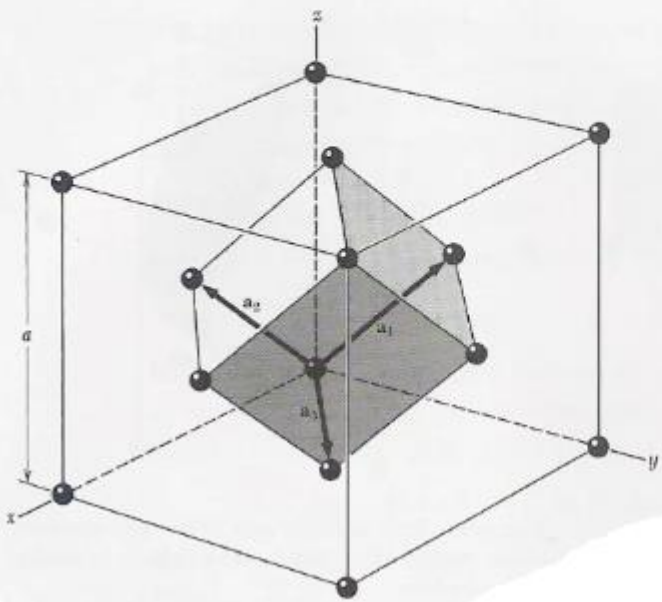
Red recíproca de la red bcc





$$\mathbf{a}_1 = \frac{1}{2}a(-\mathbf{x}+\mathbf{y}+\mathbf{z}) \quad \mathbf{a}_2 = \frac{1}{2}a(\mathbf{x}-\mathbf{y}+\mathbf{z}) \quad \mathbf{a}_2 = \frac{1}{2}a(\mathbf{x}+\mathbf{y}-\mathbf{z})$$

Red recíproca de una red fcc



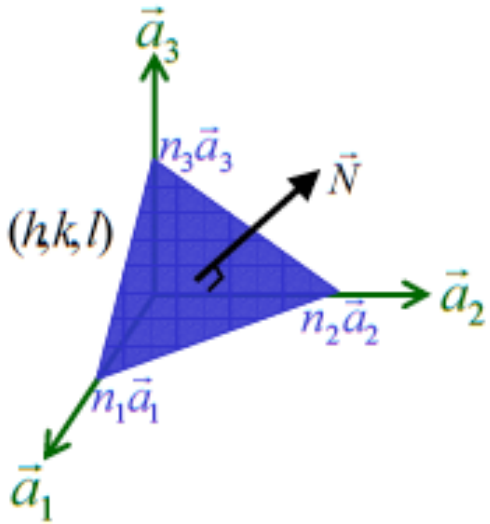
$$\mathbf{a}_1 = \frac{1}{2}a(\mathbf{y}+\mathbf{z})$$

$$\mathbf{a}_2 = \frac{1}{2}a(\mathbf{x}+\mathbf{z})$$

$$\mathbf{a}_3 = \frac{1}{2}a(\mathbf{x}+\mathbf{y})$$

Propiedades de la Red Recíproca

- **A todo plano reticular (*h k l*) le corresponde un vector del espacio recíproco, de componentes proporcionales a los índices *h*, *k*, *l*.**



$$\vec{N} = q(n_1\vec{a}_1 - n_3\vec{a}_3)\times(n_2\vec{a}_2 - n_1\vec{a}_1)$$

$$= q(n_1 n_2 \vec{a}_1\times\vec{a}_2 - n_3 n_2 \vec{a}_3\times\vec{a}_2 + n_3 n_1 \vec{a}_3\times\vec{a}_1)$$

$$= \frac{qV_0}{2\pi}(n_1 n_2 \vec{b}_3 + n_3 n_2 \vec{b}_1 + n_3 n_1 \vec{b}_2)$$

$$= \frac{qV_0 n_1 n_2 n_3}{2\pi}\left(\frac{\vec{b}_1}{n_1} + \frac{\vec{b}_2}{n_2} + \frac{\vec{b}_3}{n_3}\right)$$

$$h = \frac{1}{n_1}$$

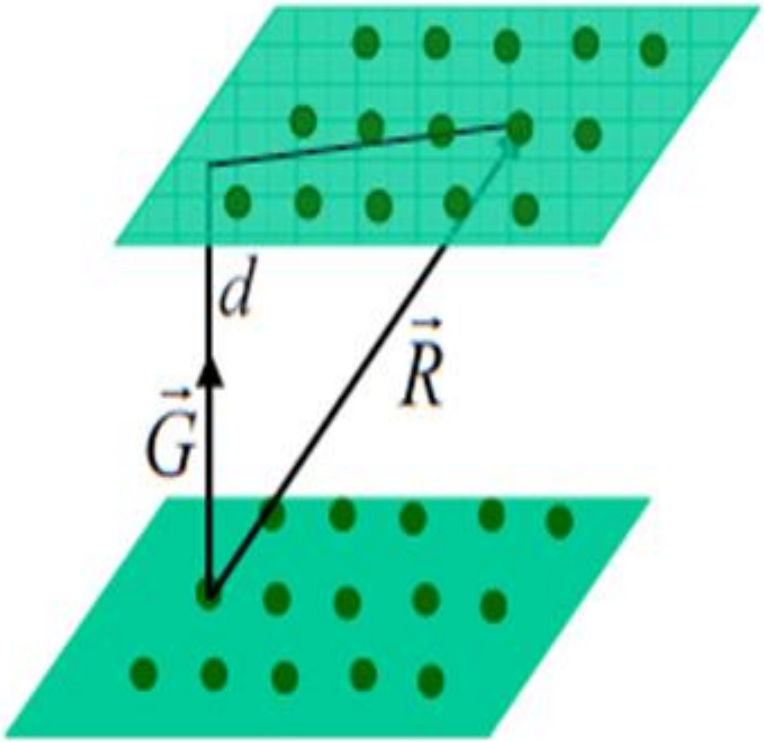
$$k = \frac{1}{n_2}$$

$$l = \frac{1}{n_3}$$

con $\quad q = \dfrac{2\pi}{V_0 n_1 n_2 n_3}$ se tiene

$$\vec{G} = h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3$$ (vector recíproco normal al plano)

El espaciado entre planos reticulares cristalinos viene dado por la relación

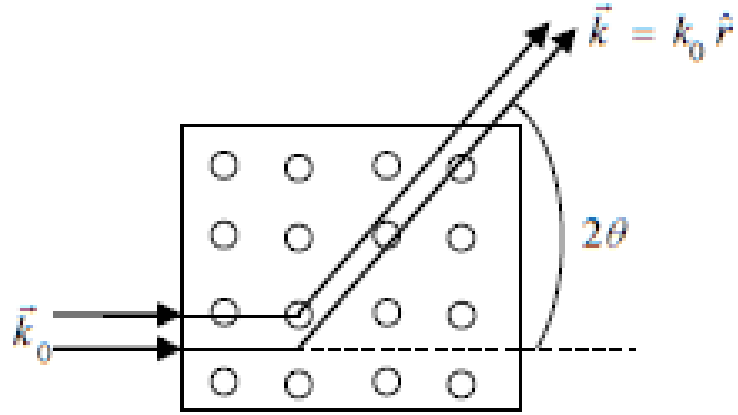


$$d_{hkl} = \frac{2\pi}{\left|\vec{G}(h,k,l)\right|}$$

Teoría del scattering desde cristales

La forma de una onda dispersada que interactúa con un solo átomo en el origen está dada, lejos del átomo, por

$$\psi \approx A e^{-i\omega t}\left[e^{i\vec{k}_0\cdot\vec{r}} + f(\hat{r})\frac{e^{ik_0 r}}{r} \right]$$



Factor de forma atómica

(contiene detalles de la interacción entre el potencial de scattering y la onda dispersada)

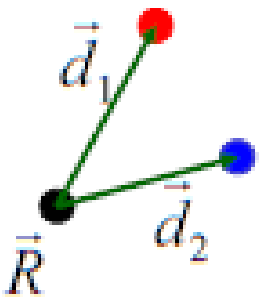
$$I_a \equiv \frac{d\sigma}{d\Omega_{atom}} = \left| f(\hat{r}) \right|^2$$

Para scattering de rayos X, el potencial es proporcional a la densidad electrónica:

$$f(\vec{k}) = -\frac{1}{e}\int d^3r\, e^{i\vec{k}\cdot\vec{r}}\,\rho(\vec{r})$$

razón de la amplitud dispersada por un átomo y la de un electrón aislado

Scattering desde una red con base



$$\text{Amplitud total} \propto \sum_{\vec{n}} \sum_{j=1}^{m} f_j e^{-i\vec{q}\cdot(\vec{R}_{\vec{n}}+\vec{d}_j)}$$

$$= \sum_{\vec{n}} e^{-i\vec{q}\cdot\vec{R}_{\vec{n}}} \sum_{j=1}^{m} f_j e^{-i\vec{q}\cdot\vec{d}_j}$$

Factor de estructura geométrica:

$$S_{\vec{G}} = \sum_{j=1}^{m} f_j e^{-i\vec{G}\cdot\vec{d}_j}$$

$$\vec{G} = h\overrightarrow{b_1}+k\overrightarrow{b_2}+l\overrightarrow{b_3}$$

$$\overrightarrow{d_j} = x_j\overrightarrow{a_1}+y_j\overrightarrow{a_2}+z_j\overrightarrow{a_3}$$

$$\vec{G}.\overrightarrow{d_j} = (h\overrightarrow{b_1}+k\overrightarrow{b_2}+l\overrightarrow{b_3}).(x_j\overrightarrow{a_1}+y_j\overrightarrow{a_2}+z_j\overrightarrow{a_3})$$

$$\vec{b}_i \cdot \vec{a}_j = 2\pi\,\delta_{ij}$$

$$\vec{G}.\overrightarrow{d_j} = 2\pi(hx_j+ky_j+lz_j)$$

$$S_{\vec{G}} = \sum_{j=1}^{m} f_j e^{-i\vec{G}\cdot\vec{d}_j} = \sum_{j}^{n} f_j\, e^{-2\pi i(hx_j+ky_j+lz_j)}$$

FACTOR DE ESTRUCTURA DE LA BCC

El factor de estructura solo se aplica a las bases. La BCC se puede considerar como una base con átomos idénticos con $\vec{r}=0$ y $\frac{a}{2}(\hat{x},\hat{y},\hat{z})$; $a=1$.

$$S_G = f e^{0} + f e^{-i2\pi\left(h\frac{1}{2}+k\frac{1}{2}+l\frac{1}{2}\right)}$$

$$S_G = f + f e^{-i\pi(h+k+l)} = f\left[1+e^{-i\pi(h+k+l)}\right]$$

Cuando
$$\begin{cases} h+k+l = \# \text{ impar} \longrightarrow S=0 \\ h+k+l = \# \text{ par} \longrightarrow S=2f \end{cases}$$

FACTOR DE ESTRUCTURA DE LA FCC

La base de la FCC tiene átomos idénticos en $\vec{r}=0$, $(0,\frac{1}{2},\frac{1}{2})$, $(\frac{1}{2},0,\frac{1}{2})$ y $(\frac{1}{2},\frac{1}{2},0)$

$$S_G = fe^{0} + fe^{-i\pi(k+l)} + fe^{-i\pi(h+l)} + fe^{-i\pi(h+k)}$$

$$S_G = f\left[1 + e^{-i\pi(k+l)} + e^{-i\pi(h+l)} + e^{-i\pi(h+k)}\right]$$

Si todos los índices son enteros pares $\longrightarrow$ $S=4f$.

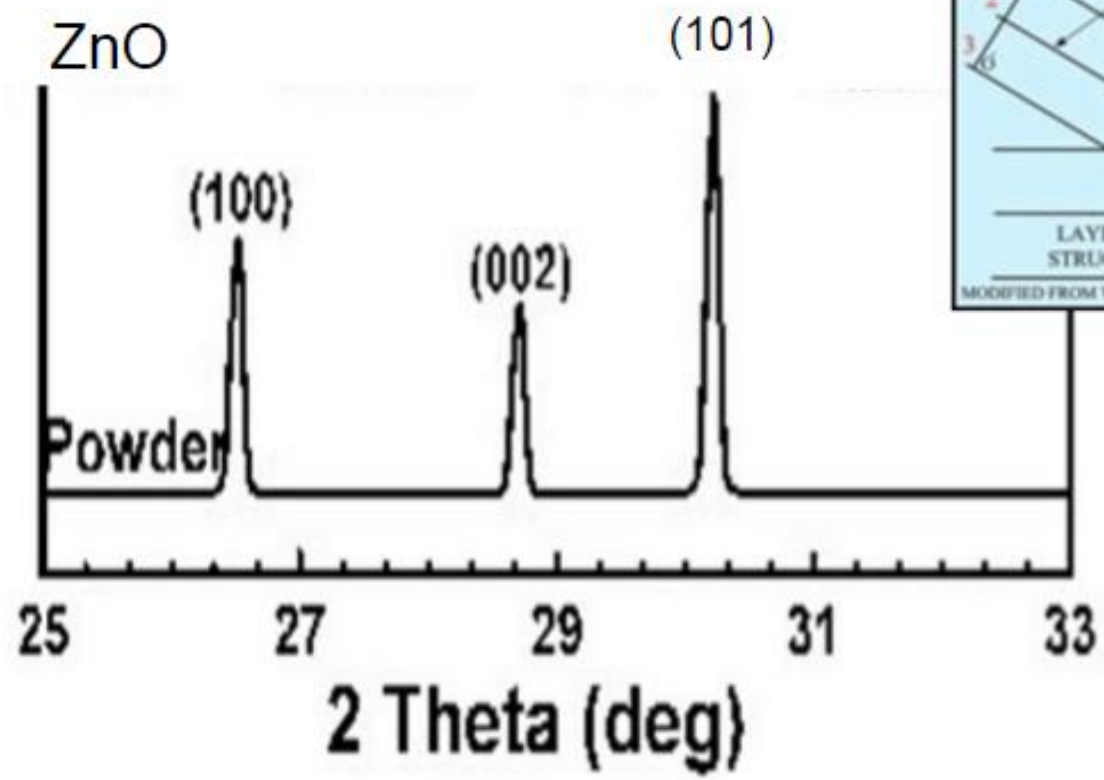
Si todos los índices son impares $\longrightarrow$ $S=4f$.

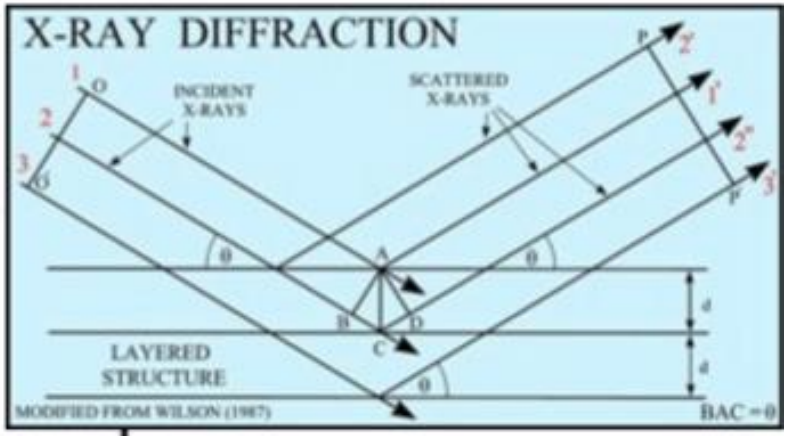
Si sólo uno de los índices es par $\longrightarrow$ $S=0$

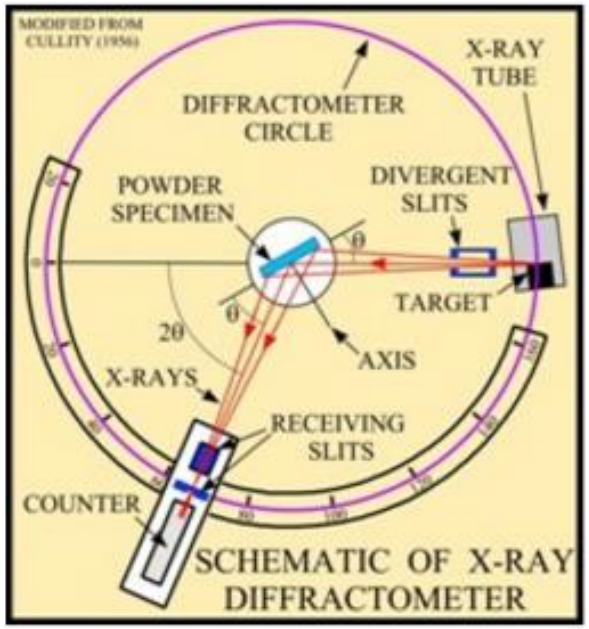
Si sólo uno es impar $\longrightarrow$ $S=0$

Difracción de Rayos X

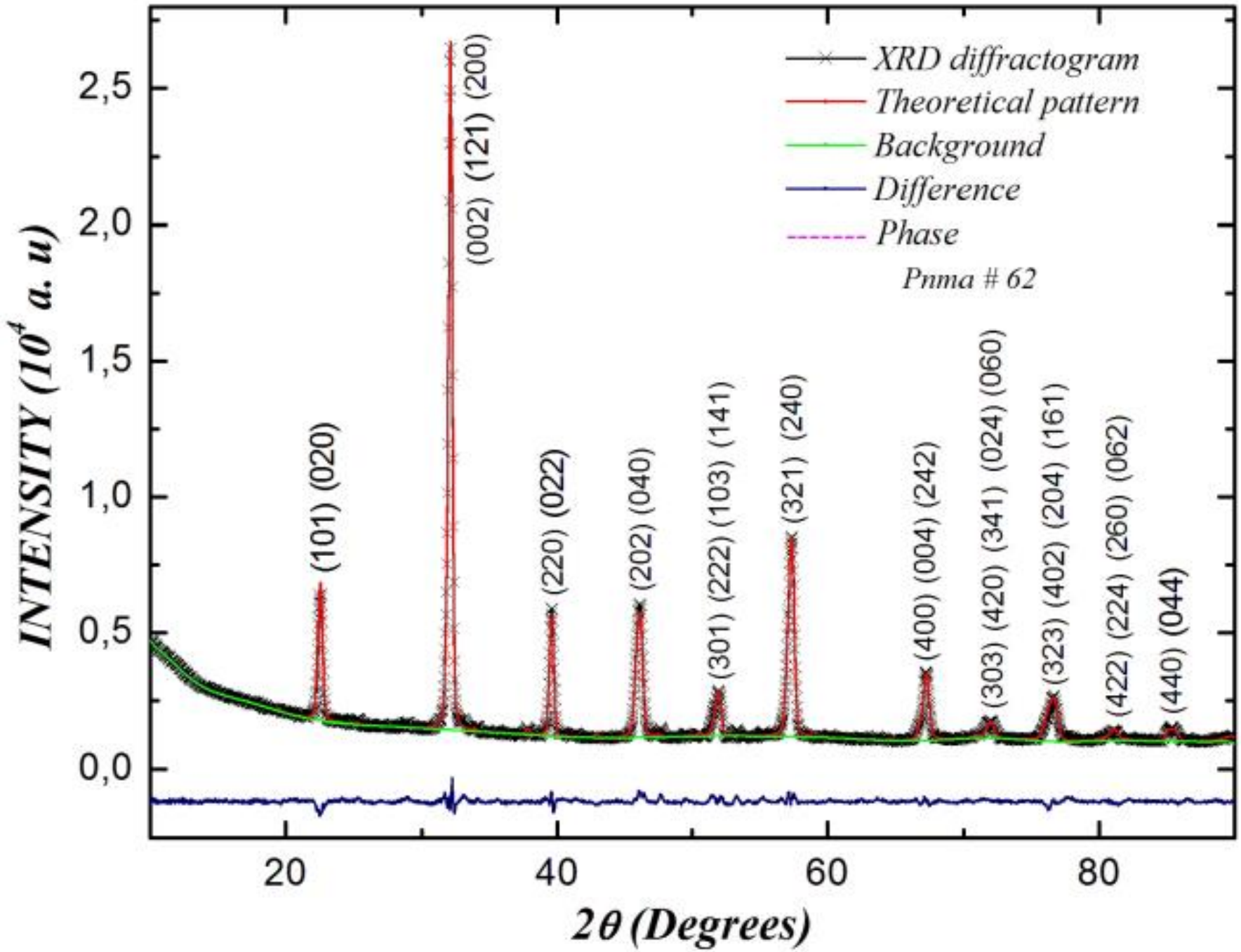
Los planos cristalinos difractan

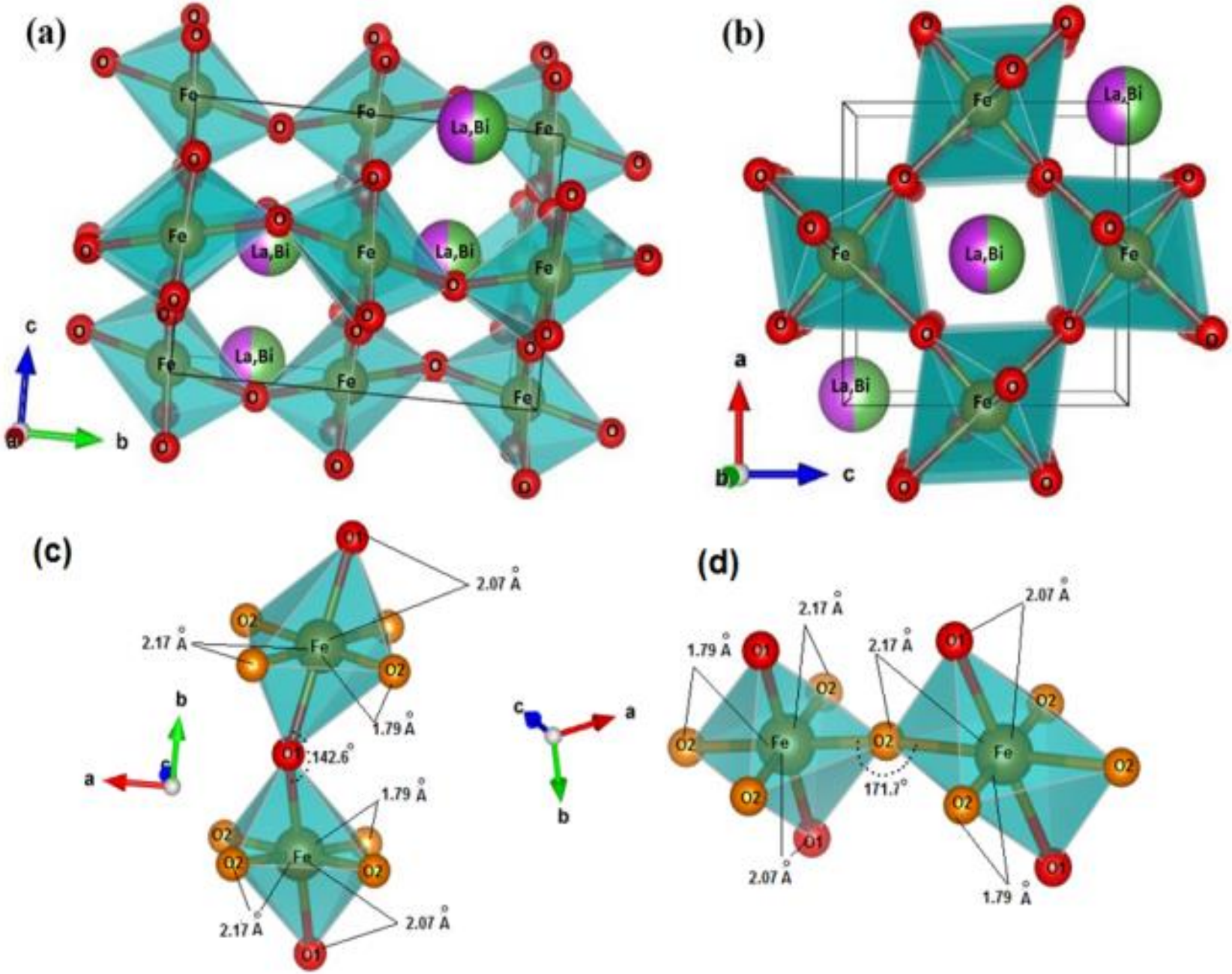






- Posición de la línea: identificación
  - Ancho de la línea: tamaño


XRD diffractogram
Theoretical pattern
Background
Difference
Phase
Pnma # 62
(101) (020)
(002) (121) (200)
(220) (022)
(202) (040)
(301) (222) (103) (141)
(321) (240)
(400) (004) (242)
(303) (420) (341) (024) (060)
(323) (402) (204) (161)
(422) (224) (260) (062)
(440) (044)
INTENSITY (10^4 a. u)
2θ (Degrees)
2,5
2,0
1,5
1,0
0,5
0,0
20
40
60
80


(a)
Fe
La,Bi
O
c
b
a
(b)
Fe
La,Bi
O
a
b
c
(c)
O1
O2
Fe
2.07 Å
2.17 Å
1.79 Å
142.6°
(d)
2.17 Å
2.07 Å
1.79 Å
171.7°
O1
O2
Fe

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Synthesis process and structural characterization of the Sr_2EuRuO_6 complex perovskite

C.A. Triana, D.A. Landínez Téllez, J. Roa-Rojas*

Grupo de Física de Nuevos Materiales (GFNM), Departamento de Física, Universidad Nacional de Colombia, Bogotá D.C. A.A. 5997, Colombia

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ABSTRACT

The Sr_2EuRuO_6 complex perovskite has been synthesized by the solid-state reaction method and the crystal structure, surface morphology and composition have been investigated. Results of powder X-ray diffraction measurements and Rietveld analysis show that this compound crystallizes in a monoclinic distorted perovskite-type structure, which belongs to the monoclinic $P2_1/n$ (#14) space group, that corresponds to the $(a^+b^-b^-)$ tilt system on the Glazer notation. The structure presents an alternating distribution of the Ru^{5+} and Eu^{3+} ions on the six coordinate M sites, while the Sr^{2+} is located in the A-site of the Sr_2EuRuO_6 complex perovskite, with lattice parameters $a = 5.7996(5)$ Å, $b = 5.8960(7)$ Å, $c = 8.3234(6)$ Å, angle $\beta = 90.234(7)^\circ$ and $V = 284.61(4)$ Å^3. Morphological analysis of this material, performed by scanning electron microscopy (SEM), allows to establish the granular feature of compound with agglomerates from amongst ≈1 to 3 μm size, and by means of the Scherrer formula was calculated a particle size of $D = 34.2$ nm. Result suggests that crystal structure of the Sr_2EuRuO_6 suffers grain size-induced polarization rotation, which produces a phase transition toward monoclinic phase $P2_1/n$ (#14).

3.1. Structural results

Fig. 1 shows the powder X-ray diffraction patterns for the Sr_2EuRuO_6 complex perovskite after of annealing treatments at

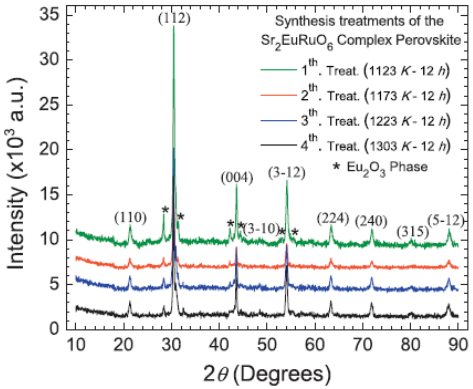

Fig. 1. X-ray diffraction pattern for the Sr_2EuRuO_6 complex perovskite after of calcined treatments at 1123 K, 1173 K, 1223 K and 1303 K, respectively, for a same time interval of time of 12 h.

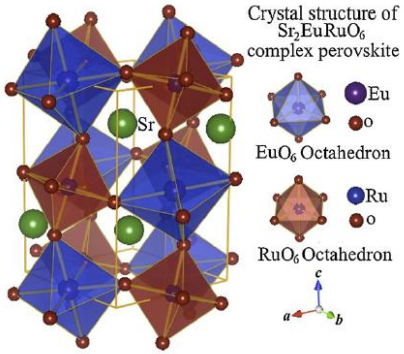

Fig. 3. Crystal structure of the Sr_2EuRuO_6 complex perovskite belong to the $P2_1/n$ (#14) space group. The cube in continuous line indicates the primitive unit cell a_p.

which predicts 18.31° and
octahedral occurs because t
strained to tilt so as to optim
since the Ru^{5+} cation is small
inter atomic bond lengths E
octahedral are shown. One
are lightly longer than equa
of the $EuO_{6/2}$ and $RuO_{6/2}$ o
bond lengths, main inter-at
structural parameters obtai
X-ray diffraction patterns b
are shown in Table 1.

3.2. Morphological results

The agglomerates and th
the study of the crystal struc
complex perovskite, since t
the size distribution sets th
of the compound. In additi
the ceramic sample strong

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Magnetic and structural properties of the new double perovskite family $Sr_2GdRu_{1-x}Re_xO_y$

L. T. Corredor,[1,2,a)] J. Roa-Rojas,[2] D. A. Landínez Téllez,[2] R. Beltrán,[2] P. Pureur,[3] F. Mesquita,[3] and J. Albino Aguiar[1]

[1]*Departamento de Física, Universidade Federal de Pernambuco, 50670-901 Recife-PE, Brazil*
[2]*Grupo de Física de Nuevos Materiales, Departamento de Física, Universidad Nacional de Colombia, Bogotá D.C., Colombia*
[3]*Instituto de Física, Universidade Federal do Rio Grande do Sul, Porto Alegre-RS, Brazil*

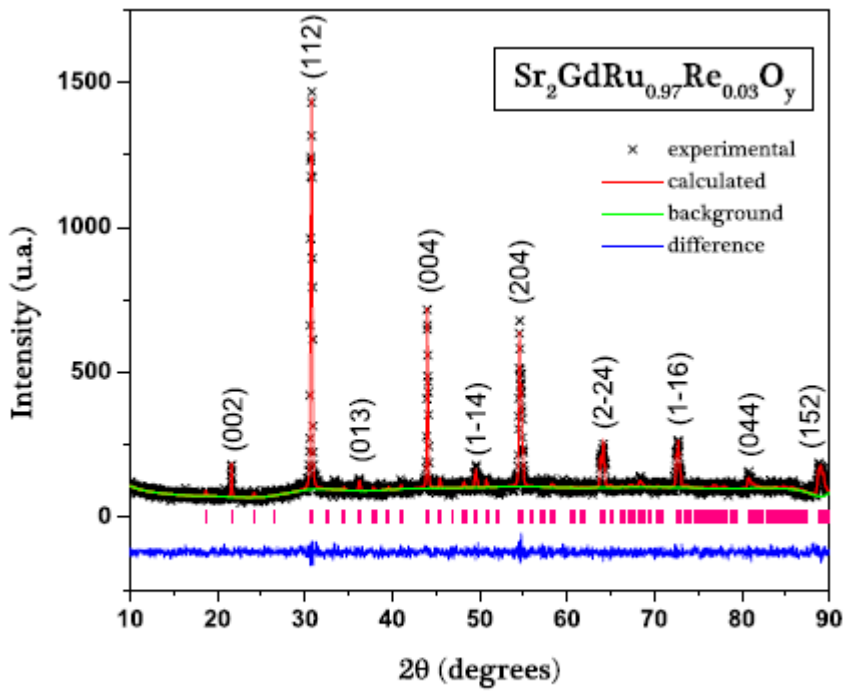

FIG. 1. Rietveld refinement of the X-ray diffraction pattern for the sample $Sr_2GdRu_{1-x}Re_xO_y$ with $x = 0.03$.

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Ferroelectric, magnetic and structural studies of the $Bi_4LaSmFe_2Ti_3O_{18}$ multiferroic material

C.E. Alarcón-Suesca [a,b,*], J.A. Cardona-Vásquez [b], J.P. Salcedo-Fontecha [c], A. Vargas-Jiménez [c], D.A. Landínez-Téllez [b], J. Roa-Rojas [b]

[a] Fachgebiet Synthese und Charakterisierung Innovatiert Materialien, Chemistry Department, Technische Universität München, Lichtenbergstrasse D-85748, Garching, Germany

[b] Grupo de Física de Nuevos Materiales, Departamento de Física, Universidad Nacional de Colombia, AA 5997 Bogotá DC, Colombia

[c] Maestría en Ciencias Física, Departamento de Física, Universidad Nacional de Colombia, AA 5997 Bogotá DC, Colombia

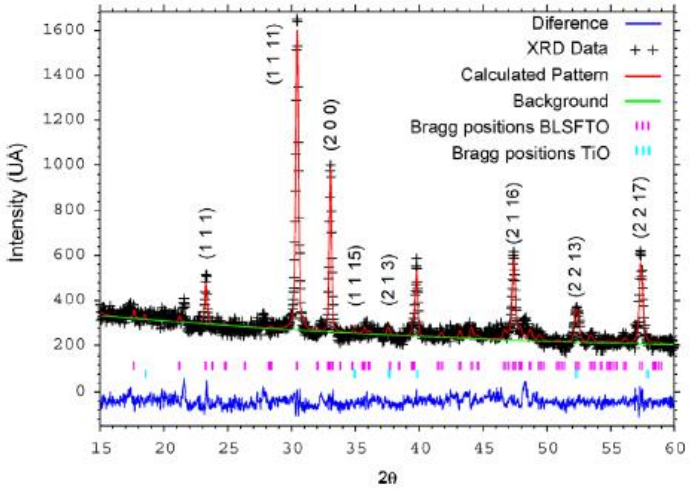

Fig. 1. XRD pattern of Aurivillius BSLTFO main phase. Bragg positions for Aurivillius oxide are noted below. Difference between simulated and experimental data is drawn in blue.

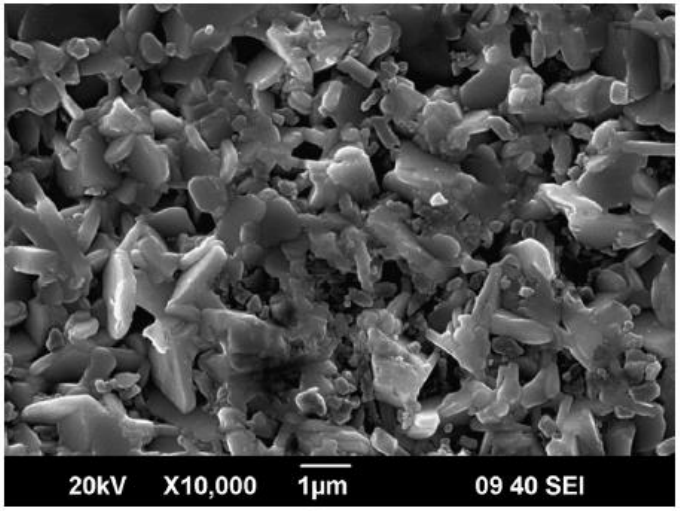

Fig. 2. SEM image of final BLSFTO sample sintered at 900 °C. Formation of plate-like grains is characteristic of Aurivillius oxides.

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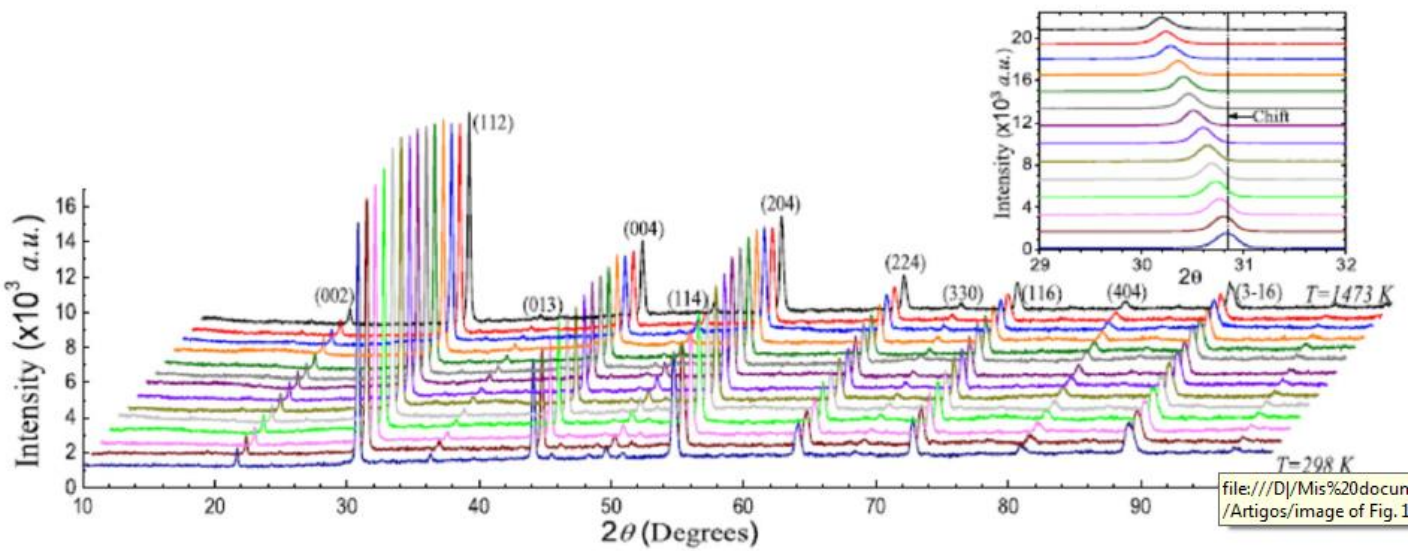
General study on the crystal, electronic and band structures, the morphological characterization, and the magnetic properties of the Sr_2DyRuO_6 complex perovskite

C.A. Triana, D.A. Landínez Téllez, J. Roa-Rojas *

Grupo de Física de Nuevos Materiales, Departamento de Física, Universidad Nacional de Colombia, Bogotá D.C., AA 5997, Colombia



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Chemical Stability and Crystallographic Analysis of the the Sr_2HoNbO_6 Cubic Perovskite as Potential Substrate for $YBa_2Cu_3O_{7-\delta}$ Superconducting Films

Randirley Beltrán Rodríguez[a], David Landínez Téllez[a], Jairo Roa-Rojas[a]*

[a]Grupo de Física de Nuevos Materiales, Departamento de Física, Universidad Nacional de Colombia, AA 5997, Bogotá DC, Colombia

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The Sr_2HoNbO_6 complex perovskite has been synthesized by solid-state reaction. X-ray diffraction pattern reveals that Sr_2HoNbO_6 material crystallizes in a cubic structure, $Fm\bar{3}m$ space group, lattice

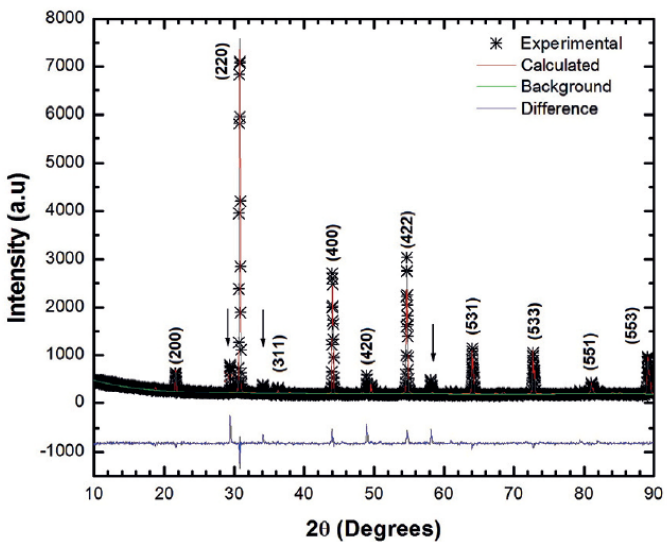

Figure 1: Characteristic XRD pattern for the complex perovskite Sr_2HoNbO_6. Symbols represent experimental diffraction data and base line is the difference between experimental and simulated

The lattice parameters of SHNO, calculated from the XRD d are *a=8,018(0) Å* while for the YBCO sample we obtair *a=3,866(1) Å, b=3,874(2) Å* and *c=11,651(9)* Å. The structu parameters experimentally obtained are shown in table 1. the other hand, it is observed in figure 1 the presence of th peaks, which do not belong to the SHNO. From the Rietv refinement it was concluded that the rare-earth precur oxide Ho_2O_3 is present in the sample in a percentage of 0,8 Arrows in figure 1 signal the respective peaks.

Figure 2 shows the structure of the SHNO doul perovskite constructed from the data of Rietveld refineme

As expected from the discussion above, the catio ordering of the Ho^{3+} and Nb^{5+} in the B and B' sites of cubic double perovskite is clearly observed. On the otl hand, no tilting or rotation of the Ho-O_6 and Nb-O_6 octahe is observed in the perfect cubic structure.

The magnetic response of SHNO has been investigated measuring the DC magnetic susceptibility in the temperat range *50* to *300 K* and at an applied magnetic field of *5*

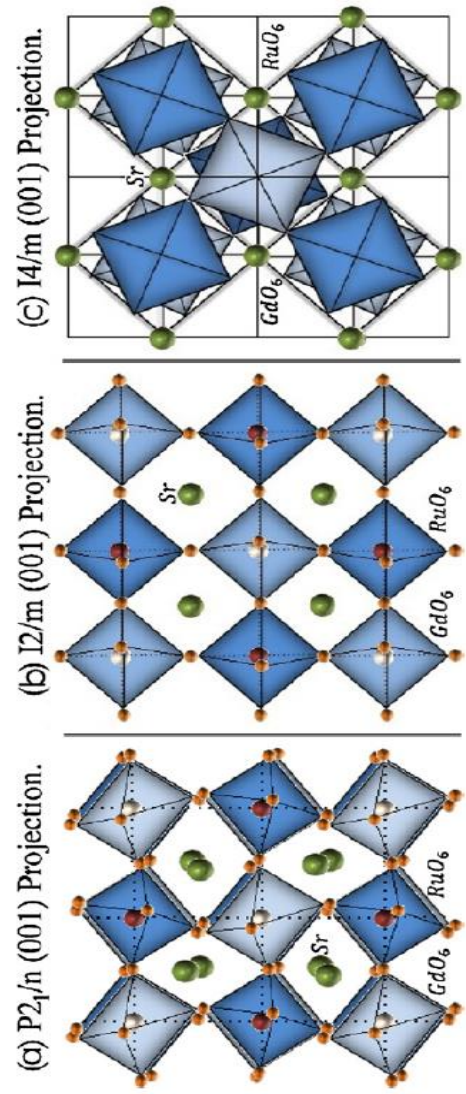

Fig. 7. (0 0 1) Projections of monoclinic $P2_1/n$ (#14) (*a*), monoclinic I2/*m* (#12) (*b*) and tetragonal I4/*m* (#87) (*c*) phases of Sr_2GdRuO_6 complex perovskite.